USTAD

See inside any machine.
Know what's failing. Fix it right.

AI diagnosis on Meta glasses that reveals the hidden failing part, proves the cause, and coaches the repair, with a built-in tutor that turns every expert into a teacher.

Meta Wearables Studio UAE 2026 · Theme: Education
Team Ustad: Shabeer Ahamed Kamal (Team Lead, AI & Computer Vision) · Keval Doshi (AI & App Developer)

1. The idea in one page

Ustad = what's going on inside the machine, why, and how to fix it.

The problem. When a machine fails, people only see the outside: a leak, a noise, smoke, an error code. The real cause is hidden inside. Only experienced technicians know which internal part is failing, and they are rare, expensive and retiring. Junior technicians guess, replace the wrong parts and repeat visits. Manuals are useless when your hands are full of tools.

The solution. A technician looks at a machine through Meta glasses. Ustad reads the outside clues (what it sees and hears), works out which inside part is failing, **reveals it in an exploded view** with the failing part glowing red, explains the cause chain, gives a 10-second test to **prove** the diagnosis, and then **coaches the repair** step by step in the technician's own language.

The add-on. Ustad Tutor lets an expert record a repair once through the glasses. AI turns it into a lesson, and every new technician learns it hands-free. Every confirmed fix makes the system smarter.

What others do	What Ustad does
Phone apps guess the fault from a photo (FixBot, Repflow, FixMode)	Hands-free on glasses, while you work
Text answers	Shows the **inside** of the machine, with the failing part highlighted
Camera only	**Sight + sound** (+ optional sensor or thermal data)
One guess	Ranked causes + a **confirmation test** before you touch anything
Fix instructions	Live coaching + expert lessons + skill certification

2. How it works

Step	What happens	Example (AC leaking water)
1. LOOK	Spots symptoms: leaks, smoke, rust, cracks, burn marks, error codes, warning lights. Reads the nameplate to identify the exact make and model.	Sees water dripping under the indoor unit. Reads 'Split AC, 1.5 ton'.
2. LISTEN	Hears abnormal sounds through the glasses mic: grinding bearings, struggling compressor, pump cavitation, rattling fans.	Fan sounds normal, compressor running.
3. REVEAL	The machine explodes into its parts on the Ray-Ban Display (or phone). The failing part glows red, with the cause chain.	Drain pan + drain pipe glow red. 'Drain line clogged → water backs up → overflow leak.'
4. CONFIRM	Ranks likely causes with confidence and gives a quick safe test to prove the top one.	'Pour a cup of water into the tray. If it doesn't drain in 10 s: confirmed.'
5. FIX	Step-by-step voice coaching, parts list, photo checkpoints. Safety Stop: tells you when to stop and call a certified pro.	'Power off. Remove the front panel. Flush the drain line…' + checkpoint photo.
6. LEARN	Saves the confirmed fix to the machine's history and the shared knowledge base, so diagnosis gets better for everyone.	Next time this model leaks, drain line is ranked #1 instantly.

Ustad Tutor (add-on)

- **Record:** an expert says "Hey Ustad, start lesson" and performs a repair while talking naturally.
- **Build:** AI turns it into steps, tips, safety warnings and photo checkpoints. The expert approves it on their phone.
- **Teach:** trainees hear the lesson in their own language (Arabic, English, Hindi, Urdu, Malayalam, Tagalog…).
- **Ask the Ustad:** "Why do we do this?" is answered in the expert's own recorded words.
- **Certify:** pass all checkpoints and earn a digital skill badge. Supervisors see progress on a dashboard.

3. Feature list

Feature	Description
Machine ID	Reads nameplates, labels, QR codes and model numbers; loads the right 3D model and manual.
Symptom Vision	Detects leaks, smoke, corrosion, cracks, loose parts, burn marks, error codes and indicator lights.
Machine Ears	Sound analysis of the running machine to detect mechanical faults.
X-Ray Reveal	Exploded view of the machine with the failing part highlighted and labelled.
Cause Chain	Explains why, in plain words: root cause → effect → symptom.
Prove-It Test	Ranked diagnosis with confidence %, plus a quick, safe confirmation test.
Repair Coach	Hands-free step-by-step guidance with photo checkpoints.
Safety Stop	Blocks dangerous shortcuts (power, gas, refrigerant, high voltage) and escalates to a pro.
Any Language	Voice in and out in the technician's language.
Live Expert Call	One voice command connects a senior expert who sees what you see.
Machine History	Every visit, diagnosis and fix saved per machine.
Auto Report	Job report with photos, cause and parts used, generated automatically for the manager.
Ustad Tutor	Expert-recorded lessons, trainee coaching and skill badges (add-on).

Modules

Module	Status	Data / source
Split AC units (leaks, icing, noise, not cooling)	MVP: GITEX demo	Manufacturer manuals, curated fault tree, 3D model
Lithium-ion batteries (health, overheating, swelling)	MVP: 2nd demo	NASA Li-ion Battery Aging Dataset + synthetic data
Pumps & motors (bearings, cavitation, vibration)	Next	Public machine-sound fault datasets
Solar inverters, EV parts, appliances	Later	Partner manuals and field data

4. How we build it

Layer	Choice
Glasses	Ray-Ban Meta (camera, mic, speakers). Ray-Ban Display for the in-lens exploded view, with phone or tablet view as fallback.
Glasses SDK	Meta Wearables Device Access Toolkit 1.0 (native iOS/Android) + Display web apps. Develop with the Mock Device Kit.
Phone app	Flutter or native companion app: camera frames, audio, voice commands, 3D viewer.
Vision + reasoning	Multimodal LLM (Claude) for symptom detection, diagnosis, cause chain and coaching.
Sound	Audio classifier for machine faults (spectrogram + CNN), trained on public datasets.
Battery model	Health and remaining-life prediction trained on the NASA battery dataset.
3D Reveal	Pre-built, part-labelled 3D models (GLB) per machine type; exploded animation rendered to images/video for the glasses.
Voice	Speech-to-text (incl. Arabic) + translation + text-to-speech.
Backend	Knowledge base (manuals, fault trees, confirmed fixes), machine history, dashboard. UAE-region cloud.

Honest limits we design around

- The glasses can't literally see inside. Ustad reasons from outside clues, like an expert does, and always shows confidence and a confirmation test.
- A normal camera can't see heat. For heat faults we add sensor data or an optional phone thermal camera.
- Exploded views come from a curated 3D model library per machine type, not generated from a single photo.
- Privacy: recording only on voice command, consent built in, bystander faces blurred, data kept in a UAE region.

Build plan

When	Deliverable
Week 1	Glasses ↔ phone connection, Machine ID, Symptom Vision on the AC unit
Week 2	X-Ray Reveal (3D AC model + exploded animation), Cause Chain, Prove-It Test
Week 3	Repair Coach, multilingual voice, Safety Stop, battery module (NASA)
Week 4	Ustad Tutor add-on, dashboard, demo rehearsal + backup video

5. Market, demo and pitch

Who pays

- **Facilities management & AC service companies:** huge multilingual technician workforces; AC is critical in the UAE.
- **Energy & battery storage:** Masdar's 19 GWh battery project in Abu Dhabi and 8 GW of planned storage need inspection at scale.
- **EV workshops and e-mobility fleets:** battery checks for EVs, e-bikes and e-scooters (real battery fires in Dubai).
- **Vocational schools & training centres:** Ustad Tutor for hands-on skills.

Business model: monthly subscription per technician, plus paid modules per machine type and an enterprise Expert Library.

Timing

Microsoft stopped selling Dynamics 365 Guides and Remote Assist, which reach end of support on 31 December 2026, and HoloLens 2 is discontinued. Companies need a replacement, and Meta just opened the Ray-Ban Display screen to developers (May 2026).

GITEX demo (3 minutes)

- A split AC unit on stage is dripping water.
- The technician looks at it. Ustad identifies the unit, spots the leak and listens to it run.
- **Reveal:** the AC explodes into parts in the glasses; the drain pan and pipe glow red.
- "Drain line clogged, 87% confidence. Pour water in the tray to confirm." → Confirmed.
- Ustad coaches the fix in English, then a judge hears the same steps in Hindi.
- Bonus: hold up a battery pack and Ustad reveals the weak cell inside (NASA model).

"Ustad: see inside any machine. Know what's failing. Fix it right."

6. AI prompts

The core system prompts that power Ustad. Each agent returns structured JSON so the app can drive the glasses display, voice and 3D view.

Prompt 1: Look & Listen (symptom detection)

```
You are Ustad's inspection eye. You receive camera frames from smart glasses
and an optional audio description of the running machine.

Tasks:
1. Identify the machine: type, brand, model (read nameplates, labels, QR codes).
2. List every visible or audible symptom: leaks, drips, stains, smoke, corrosion,
   cracks, burn marks, loose parts, error codes, indicator lights, abnormal sounds.
3. For each symptom give its location on the machine and how sure you are.
Rules: only report what is actually visible or audible. Never invent symptoms.
If the image is unclear, ask the user to move closer or change the angle.

Return JSON:
{ "machine": {"type": "", "brand": "", "model": "", "confidence": 0.0},
  "symptoms": [{"symptom": "", "location": "", "evidence": "", "confidence": 0.0}],
  "need_better_view": false, "user_instruction": "" }
```

Prompt 2: Diagnose & Reveal (find the hidden failing part)

```
You are Ustad, a master technician with 30 years of experience.
Input: machine identity, symptoms (JSON), and the machine's part list
from the 3D model library: [{"part_id": "", "name": ""}, ...].

Reason like an expert: map outside symptoms to the internal parts that cause them.
1. Rank the 3 most likely root causes, each tied to a part_id from the list.
2. For each, give the cause chain: root cause -> effect -> visible symptom.
3. Give a quick, SAFE confirmation test the technician can do in under 1 minute.
4. Flag any safety hazard (electricity, gas, refrigerant, high pressure, battery).

Return JSON:
{ "diagnoses": [{"rank": 1, "part_id": "", "part_name": "", "confidence": 0.0,
                 "cause_chain": ["", "", ""], "confirm_test": "",
                 "expected_result_if_true": ""}],
  "highlight_part_ids": [""], "safety": {"hazard": "", "stop_and_call_pro": false},
  "voice_summary": "<= 25 words, spoken to the technician" }
```

Prompt 3: Confirm (prove the diagnosis)

```
The technician performed the confirmation test for diagnosis #{rank}.
Inputs: the test, the expected result, a checkpoint photo and/or the
technician's spoken description of what happened.

Decide: CONFIRMED, NOT_CONFIRMED or UNCLEAR.
If NOT_CONFIRMED, move to the next-ranked diagnosis and give its test.
If UNCLEAR, tell the technician exactly what to look at again.

Return JSON:
{ "result": "CONFIRMED", "reason": "", "next_action": "",
  "voice_summary": "<= 20 words" }
```

Prompt 4: Repair Coach (hands-free fix)

```
You are Ustad coaching a technician through a repair. Hands are busy and eyes
are on the job, so speak in short, clear steps in the technician's language:
{language}. Confirmed fault: {diagnosis}. Machine: {machine}.

Rules:
- One action per step, max 15 words. Wait for "next" before continuing.
- Always start with safety isolation (power off, valves closed, PPE).
- Mark critical steps as CHECKPOINT: ask for a photo and say what it must show.
- On "why?", explain briefly using the expert lesson notes if available.
- If anything is unsafe or beyond a field repair, say STOP and escalate.

Return JSON per step:
{ "step": 1, "say": "", "checkpoint": false, "checkpoint_must_show": "",
  "parts_or_tools": [""], "safety_stop": false }
```

Prompt 5: Ustad Tutor (turn an expert recording into a lesson)

```
Input: transcript and key frames from an expert recording a repair through
smart glasses, with timestamps.

Build a lesson:
1. Split the task into clear steps (title, instruction, timestamp of the clip).
2. Capture the expert's tips and "tricks" in their own words.
3. Mark safety-critical steps and choose 3-5 CHECKPOINT steps, each with a
   reference frame and a yes/no question an AI can verify from a photo.
4. Write a 2-sentence summary and a list of tools and parts.
Never add steps the expert did not perform. Mark anything unclear for review.

Return JSON:
{ "title": "", "summary": "", "tools": [""],
  "steps": [{"n": 1, "title": "", "instruction": "", "expert_tip": "",
             "clip_start": "00:00", "safety_critical": false,
             "checkpoint": {"enabled": false, "question": "", "reference_frame": ""}}],
  "needs_expert_review": [""] }
```

Prompt 6: Battery module (NASA data model output to explanation)

```
Input from the battery model: {"state_of_health": 0.62, "remaining_cycles": 41,
"temp_trend": "rising", "internal_resistance_trend": "rising", "cell_id": "C4"}.

Explain to a technician, simply:
1. Is this battery safe to keep using? (OK / WATCH / REPLACE / ISOLATE NOW)
2. Which internal layer is most likely degrading (anode, cathode, separator,
   electrolyte) and why, as a cause chain.
3. What to do now, in 3 short steps.
Return JSON with: verdict, highlight_layer, cause_chain, actions, voice_summary.
```

Build prompt (to give an AI coding assistant to build the MVP)

```
Build "Ustad", a companion app for Meta Ray-Ban glasses.
Stack: Flutter app + Meta Wearables Device Access Toolkit (use the Mock Device
Kit), Python FastAPI backend, Claude API for vision and reasoning.

Screens/flows:
1. Inspect: stream glasses camera frames; on "Hey Ustad, inspect" capture
   3 frames + 5 s audio, call Prompt 1, then Prompt 2.
2. Reveal: load the machine's GLB model (start with a split AC indoor unit),
   play an exploded-view animation, colour highlight_part_ids red, show labels.
   Render the key frame as an image for the Ray-Ban Display.
3. Confirm: speak confirm_test, capture photo/voice result, call Prompt 3.
4. Fix: run Prompt 4 step by step with voice "next", "why", "check".
5. Tutor: record expert session, call Prompt 5, show lesson for approval.
6. Dashboard: machine history, jobs, technician skill badges.
Requirements: all AI outputs validated as JSON; text-to-speech in the selected
language; works offline-tolerant with retries; privacy: record only on command.
```